

Data.csv file link:

<https://drive.google.com/file/d/1YUKt2xE1cxEzbzEEDMVZEMdJFIVacG3D/view?usp=sharing>

Problem 1:

Scenario:

A data scientist writes the following code:

```
from sklearn.impute import SimpleImputer
from sklearn.model_selection import train_test_split

df = pd.read_csv('data.csv')
imputer = SimpleImputer(strategy='median')
df[['Age', 'Salary']] = imputer.fit_transform(df[['Age', 'Salary']])
X = df.drop('Target', axis=1)
y = df['Target']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Question: Identify the data leakage issue. Explain why it is wrong and write the corrected code.

Problem 2:

Question: Using the provided data.csv file, write code to find:

1. Total missing values in each column
2. Percentage of missing values per column
3. Which columns have more than 50% missing values?

Problem 3:

Question: Look at the Temperature column in data.csv. It contains:

- Numeric values like 22.5, 18.6
- String 'Unknown'
- NaN values

Write code to handle this column properly before imputation.

Problem 4:

Question: For the Diagnosis column in data.csv, implement imputation with:

- Constant strategy using 'Missing' string
- Add a missing indicator column

Write the complete code using SimpleImputer.

Problem 5:

Question: Using the Age column from data.csv:

1. Calculate mean and median from training data only
2. Create two new columns: Age_mean_imputed and Age_median_imputed
3. Plot KDE for both to compare distributions
4. Which one would you choose and why?

Problem 6:

Scenario:

You have a dataset with the following columns:

Column	Missing %	Data Type	Distribution
Income	15%	Numerical	Highly skewed (right tail)
Rating	8%	Numerical	Normal (bell-shaped)
City	5%	Categorical	50 categories, one dominates (40%)
Diagnosis	22%	Categorical	3 categories (balanced: 34%, 33%, 33%)
Cabin	68%	Categorical	Many categories, none dominant

Question: For each column, recommend the BEST imputation strategy and justify your choice.